

Silicon Carbide

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Introduction

Silicon carbide (SiC) is a generic name for a material produced by numerous process routes that result in a host of different external and internal microstructures and, as a consequence, a broad range of properties. Within a SiC crystal the Si and C atoms form very strong tetrahedral covalent bonds (bond energy=4.6 eV) via sharing of electron pairs in sp^3 hybrid orbitals. The resulting properties of SiC make a substantial variety and number of applications possible, especially for use under challenging conditions. The primary driving forces for the interest in SiC for electronic applications, the principal thrust of this article, are for (1) high-power and high-temperature devices having switching speeds in the range of 5–100 kHz and resistant to radiation damage and (2) for substrates for Group IIIB nitride thin films for optoelectronic and microelectronic devices. SiC is suitable for these applications primarily because of its wide bandgap, good electrical conductivity, excellent thermal stability and conductivity and moderately close match in atomic arrangement and distances among the atoms on the (0001) basal plane with those of AlN and selected AlGaN solid solutions. It also possesses a high critical electric field strength and a high saturation electron drift velocity and can be successfully doped with donor and acceptor atoms that are ionized at room temperature. The native oxide of SiO_2 readily forms on the surfaces and can be grown thicker for select device applications. SiC device structures for high voltage (≤ 15 kV at this writing) switching applications including the classical Schottky diode (SD), junction barrier Schottky diode (JBS), the closely related merged PiN/Schottky barrier diode (MPS), as well as more complex SiC devices including the metal-oxide-semiconductor field-effect transistor (MOSFET) and the junction field-effect transistor (JFET) are commercially available. Ultra-fast switching devices including bipolar junction transistors (BJTs), insulated gate bipolar transistors (IGBTs) and thyristors are being developed.

Crystal Structures and Physical Properties

The SiC electronics field is also enhanced and made more complicated by the existence of a large number of different periodic stacking arrangements of the (0001) crystallographic planes containing atoms of silicon and carbon. In certain close-packed structures there exists a special one-dimensional polymorphism called polytypism. Polytypes are alike in the two dimensions of the close-packed planes, but differ in the stacking sequence along the direction perpendicular to these planes. The only cubic (C) SiC polytype is referred to as 3C or β -SiC, where the “3” refers to the number of silicon/carbon bilayers in the periodic stacking sequence. The hexagonal (H) wurtzite or 2H sequence also occurs in SiC. Both polytypes can also occur in more complex, intermixed forms yielding a wide range of ordered, larger period, stacked hexagonal or rhombohedral structures of which 6H and 4H are the most common and are the polytypes from which essentially all SiC devices have been fabricated. All non-cubic structures are known collectively as α -SiC. Schematics of the stacking sequences for 3C-, 4H- and 6H-SiC crystal structures are shown in Fig. 1.

A high-resolution transmission electron microscopy (TEM) image showing the atomic structure of a 3C-SiC film deposited on a 6H-SiC substrate is shown in Fig. 2. Note the periodic, chevron-like structure of the unit cells of the 6H polytype consisting of three

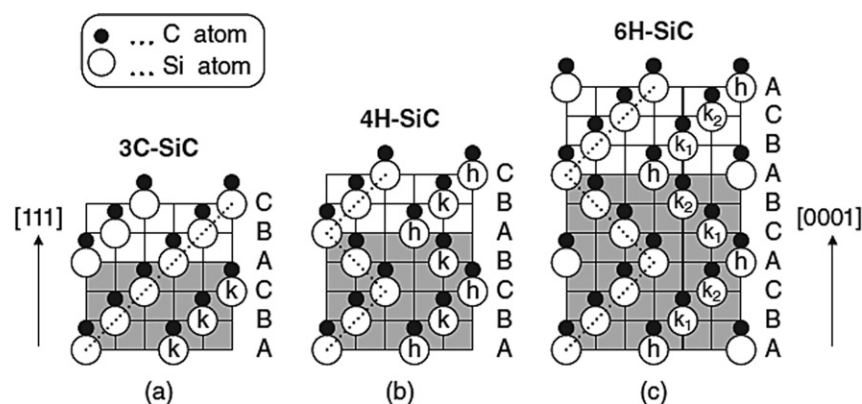


Fig. 1 Structures of the most common SiC polytypes: (a) 3C-SiC, (b) 4H-SiC, and (c) 6H-SiC. Open and closed circles denote Si and C atoms, respectively. A, B, and C represent the occupied sites in a hexagonal close-packed bilayer of Si and C (Kimoto and Cooper, 2014).

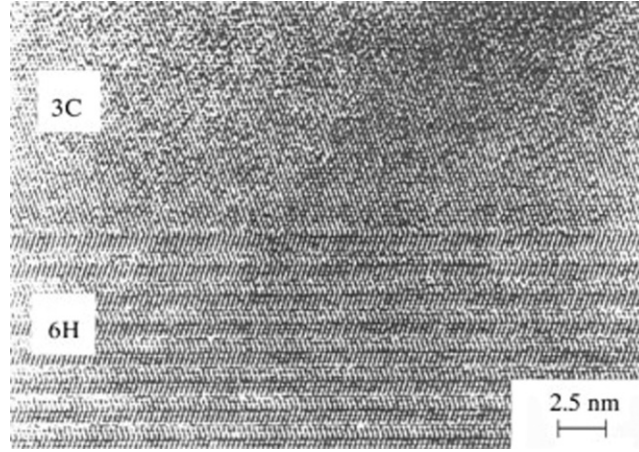


Fig. 2 High-resolution TEM image of 3C-SiC deposited on a 6H-SiC substrate (Davis *et al.*, 1991).

Table 1 Select physical properties of the 3C, 4H and 6H polytypes of SiC at room temperature

Properties/polytype	3C-SiC	4H-SiC	6H-SiC
Lattice parameters ($\text{\AA}$)	$a=4.3596$	$a=3.0798$ $c=10.0820$	$a=3.0805$ $c=15.1151$
Density (g cm^{-3})	3.21	3.21	3.21
Young's Modulus (GPa)	350–650	350–650	350–650
Poisson's ratio	0.21	0.21	0.21
Specific heat capacity ($\text{J g}^{-1} \text{K}^{-1}$)	0.69	0.69	0.69
Thermal Conductivity ($\text{W cm}^{-1} \text{K}^{-1}$)	3.3–4.9	3.3–4.9	3.3–4.9
Bandgap (eV)	2.36	3.26	3.02
Electron mobility (undoped) ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)			
$\perp$ c -axis	1000	1020	450
$\parallel$ c -axis	1000	1200	100
Hole mobility (undoped) ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	100	120	100
Electron saturated drift velocity (undoped) (cm s^{-1})	$\sim 2 \times 10^7$	2.2×10^7	1.9×10^7
Hole saturated drift velocity (undoped) (cm s^{-1})	$\sim 1.3 \times 10^7$	$\sim 1.3 \times 10^7$	$\sim 1.3 \times 10^7$
Critical (breakdown) electric field strength ($\sim 10^{15}$ electrons cm^{-3}) (MV cm^{-1})			
$\perp$ c -axis	1.4	2.2	1.7
$\parallel$ c -axis	1.4	2.8	3.0
Relative dielectric constant (100 kHz–1 MHz)			
$\perp$ c -axis	9.72	9.76	9.66
$\parallel$ c -axis	9.72	10.32	10.03

Si/C bi-layers per each half-cell with the half-cells separated by a rotational twin defect and compare with the stacking sequence shown in Fig. 1(c).

Physical Properties

The physical properties of SiC *per se* are collectively among the most extreme of any binary compound. And many of them are critical for the simulation, fabrication and operation of electronic devices. Table 1 provides the most recent data for many of these important properties for the three most common polytypes of 3C, 6H and 4H.

As a result of the different stacking sequences of the Si/C bilayers that produce the various crystal structures of the polytypes, their lattice parameters, especially the c -axis parameter, of the unit cells are very different as noted in Table 1. However, the Si–C bond energy and the Si–C bond length are essentially the same in all polytypes having the values of 4.6 eV and 1.89 $\text{\AA}$, respectively. As such, the height of the Si–C bilayer along the c -axis is approximately the same in every polytype and within the range of 2.50–2.52 $\text{\AA}$. These three nearly constant values among the polytypes ascribes very similar chemo-mechanical properties of, for example, density, specific heat, thermal conductivity, thermal expansion coefficient, Young's modulus and Poisson's ratio to these structures. By contrast, different electronic band structures and marked differences in electronic properties occur in the different

polytypes, because they possess very different periodic potentials. Thus, it is imperative that a SiC substrate and/or deposited thin film possess a single polytype structure in order that all the devices subsequently fabricated on the material exhibit similar electronic behavior under an applied electric field.

From an electronics viewpoint, the polytypes of SiC possess a range of bandgap energies at 300 K (eg, 2.36 eV (3C), 3.02 eV (6H), 3.26 eV (4H), and 3.3 eV (2H)). The value of this energy increases essentially monotonically as a function of the “hexagonality” of the polytype, which refers to the ratio of sites with hexagonal-structured surroundings of immediate atomic neighbors to the total number of Si–C bilayers containing both hexagonal- and cubic-structured sites in a unit cell. The hexagonality of the 3C, 6H, 4H and 2H polytypes is 0, 0.33, 0.5 and 1.0, respectively.

As noted in the Introduction, donor and acceptor impurities can be introduced into SiC crystals and thin films of all polytypes during growth to achieve n-type and p-type doping, respectively. By far the most commonly used n- and p-type dopants are nitrogen and aluminum, respectively. However, phosphorus and boron have also been used to produce n- and p-type doping, respectively. In all polytypes, nitrogen substitutes on the carbon sub-lattice sites; phosphorus, aluminum, and boron substitute on the silicon sub-lattice sites. The energy that must be present in the crystal lattice that is required to promote an electron from its donor energy level into the conduction band and from the top of the valence band into an acceptor energy level is referred to as the ionization energy. Interestingly, the ionization energy of the donor and acceptor impurities depends on their location in either the hexagonal- or the cubic-structure site in the crystal lattice. Values of the ionization energies of the aforementioned impurities as a function of site location, where known, and the solubility limits of these impurities are given in [Table 2](#).

As shown in [Table 1](#), the electron mobility (a.k.a. drift mobility which is defined as the drift velocity per unit of applied field) at room temperature in 4H-SiC is approximately twice that in 6H-SiC. And this difference is essentially constant regardless of the concentration of the n-type dopant. The hole mobility at room temperature is also higher in 4H up to an acceptor concentration of $\sim 2 \times 10^{19} \text{ cm}^{-3}$ where the values in 4H and 6H become equal. These data also show that the non-cubic polytypes exhibit differences (anisotropy) in mobility along the “c” and “a” axes in the crystal structure. This is particularly true for the 6H polytype.

The drift velocity of electrons and holes is the average velocity over all the electrons and/or holes that participate in the electrical conduction in the material. These velocities are proportional to the electric field strength at low values of the latter. However, at sufficiently high electric fields, these carriers increasingly interact with phonons produced by the vibrations of the crystal lattice, and the drift velocities become saturated. As the bonding between atoms is strong, the amplitudes of the lattice vibrations are small in SiC at room temperature, and the polytypes possess high values of the electron saturated drift velocity ($\sim 2 \times 10^7 \text{ cm s}^{-1}$ (3C), $1.9 \times 10^7 \text{ cm s}^{-1}$ (6H) and $2.2 \times 10^7 \text{ cm s}^{-1}$ (4H)) and the hole saturated drift velocity ($\sim 1.3 \times 10^7 \text{ cm s}^{-1}$ (all common polytypes)).

Increasing the electric field strength in the reverse-bias direction to either a Schottky barrier junction or a pn junction in a semiconductor results in an increase in the concentration of electron–hole pairs, marked increases in leakage current and the eventual breakdown of the junction. The critical or breakdown electric field strength in SiC is thus the maximum electric field that a junction can withstand. The values of this parameter parallel to the *c*-axis (the crystallographic direction of importance for high-voltage devices) in undoped 3C, 4H- and 6H-SiC are 1.4×10^6 , 2.8×10^6 and $3.0 \times 10^6 \text{ V cm}^{-1}$, respectively.

The values of thermal conductivity at 300 K of the common polytypes have the same range of 3.3–4.9 $\text{W cm}^{-1} \text{ }^\circ\text{C}^{-1}$, as shown in [Table 1](#), and are primarily an inverse function of the doping concentration. There is also a smaller dependence on the crystal direction in the non-cubic polytypes. These values are essentially the same as those of copper at room temperature.

In summary, the high electron mobilities and high critical field strengths in the SiC polytypes, especially along the *c*-axis of 4H-SiC, the availability of both donor and acceptor dopants with rather small ionization energies and a native oxide (SiO_2) gives this polytype a significant advantage over all semiconductor materials currently available for devices for high-power applications. Several reviews of the physical properties, device related studies and device development have been published ([Davis et al., 1991](#); [Ivanov and Chelnokov, 1992](#); [Choyke et al., 1997, 2004](#); [Zetterling, 2002](#); [Feng and Zhao, 2004](#); [Shur et al., 2006](#); [Friedrichs et al., 2010](#); [Kimoto and Cooper, 2014](#); [Liu et al., 2015](#)).

Additionally, the high thermal conductivity of SiC indicates the ability of very high-density integration of SiC devices. This objective is being achieved in tandem with the continued success in the scale-up of the seeded-sublimation growth technique described in the following section for producing single-crystal 4H-SiC boules from which wafers, the base material for fabrication of devices, with ever-larger diameters and low densities of defects can be obtained.

Table 2 Ionization energies and the solubility limits of nitrogen, phosphorus, aluminum, and boron in common SiC polytypes ([Kimoto and Cooper, 2014](#))

	<i>Nitrogen</i>	<i>Phosphorus</i>	<i>Aluminum</i>	<i>Boron</i>
Ionization energy (meV)				
3C-SiC	55	–	250	350
4H-SiC (hexagonal/cubic)	61/126	60/120	198/201	280
6H-SiC (hexagonal/cubic)	85/140	80/130	240	350
Solubility limit (cm^{-3})	2×10^{20}	$\sim 1 \times 10^{21}$	1×10^{21}	2×10^{19}

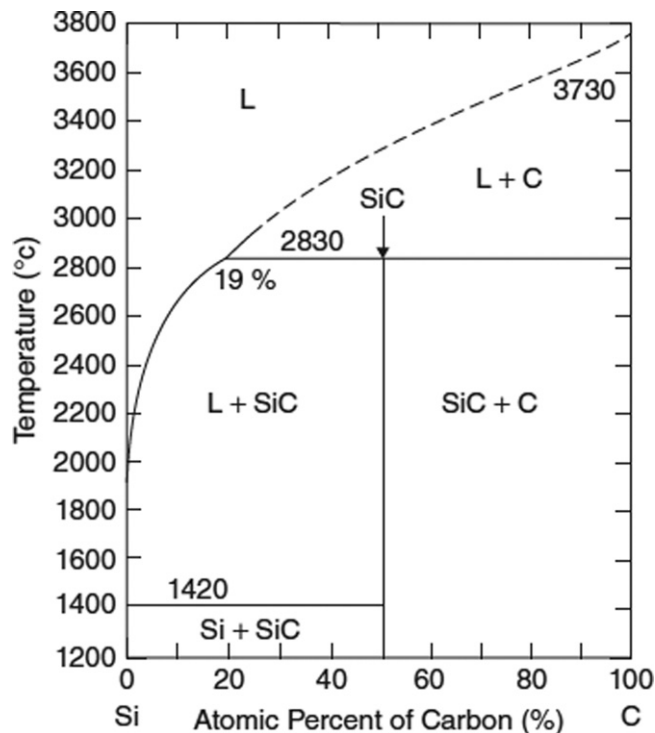


Fig. 3 Si-C phase diagram (Scace and Slack, 1959).

Growth of SiC Boules

The achievement of bulk crystal growth of compounds containing two or more elements from the periodic table relies on the knowledge of the equilibrium thermodynamics that control the chemical interactions among these components and the kinetics with which they occur. Bulk growth directly from the melt is desired; however, growth of SiC crystals is particularly challenging in this regard, as illustrated in the phase diagram in [Fig. 3](#). It does not melt congruently at 2830°C to a liquid of the same composition, but forms the two phases of solid C and a liquid containing 19 at% carbon. The silicon component in the liquid has a high vapor pressure as do the species of Si_2C and SiC_2 , and these evaporate prior to melting at atmospheric over-pressure leaving only the residual carbon. Moreover, if the Si vapor comes in contact with a hot carbon source such as graphite, it will react to form additional Si_2C and SiC_2 . In this way the individual partial pressures of Si, Si_2C and SiC_2 can be established in a closed container via control of the temperature of the environment, the amount of the SiC source material and the amount of reaction of the evaporated Si with a free carbon source. If the proper partial pressures of these three vapor species are established in the hottest part of an environmentally controlled chamber and they subsequently encounter a surface (the seed), downstream that is sufficiently cool, they will condense, diffuse on the surface, react with each other and form a SiC polytype film that, over time, can be grown vertically into a boule. The technique, referred to as either physical vapor deposition or seeded sublimation, was successfully developed for the growth of SiC boules in Russia ([Tairov and Tsvetkov, 1978, 1981](#)), from an earlier unseeded process invented in Germany ([Lely, 1955](#)).

As shown in [Fig. 4](#), a SiC source of either powder or a polycrystalline mass is placed inside a radio frequency (rf) heated graphite crucible (the free carbon source) in the bottom hot zone of the furnace, heated to cause evaporation of Si, Si_2C and SiC_2 , reaction of the Si vapor with the graphite walls and condensation of the three species on the cooler SiC seed crystal located 20–40 mm above the source. The growths are conducted at low pressure to enable rapid vapor transport from the source to the seed. The growth rate is normally in the range of 0.3–0.8 mm h⁻¹. The lengths of the boules range from 20 to 60 mm. The boules are then trepanned into wafers that are chemomechanically polished and ready for epitaxial growth of films, as discussed in the next section. The largest diameter boules available commercially at this writing are 150 mm (~6 in).

Homoeptitaxial Growth of SiC Films

Homoeptitaxial growth refers to deposition of material, most commonly as a thin film; on substrates have the same chemistry and arrangement of atoms on the crystalline surface (the same polytype in the case of SiC). SiC films are grown commercially and in most research investigations via chemical vapor deposition (CVD) from precursor sources of Si (eg, silane (SiH_4)) and C (eg, propane (C_3H_8) and ethylene (C_2H_4)) gases diluted in either pure hydrogen (H_2) or a mixture of hydrogen and argon to

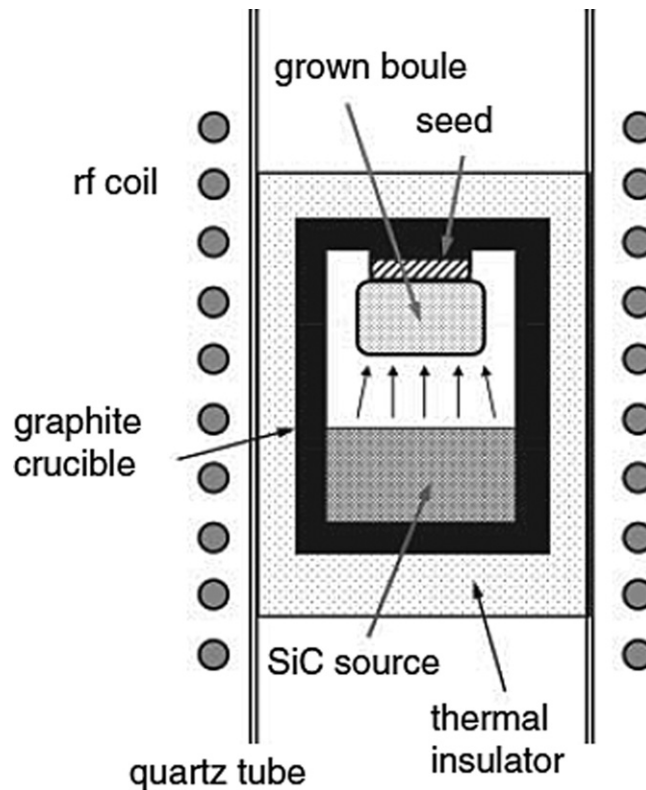


Fig. 4 Schematic of radio frequency (rf) heated sublimation equipment for seeded growth of SiC boules (Kimoto and Cooper, 2014).

prevent reactions in the gas phase. The donor and acceptor dopant sources of nitrogen (molecular nitrogen (N_2)) and aluminum (eg, trimethylaluminum) can be controllably introduced into the primary gas stream to achieve n- and p-type films. Polytype replication and films having a large range of thicknesses that are required for devices for use in low-, moderate- and high-power applications can now routinely be grown and doped on vicinal SiC {0001} substrates using various types of reactors, precursors, step-flow growth and close control of the C/Si ratio. The typical growth temperature range and rates of growth are 1500–1650°C and 3–15 $\mu\text{m h}^{-1}$, respectively. The surface of each substrate wafer is commonly etched at the growth temperature in either pure H_2 or a mixture of H_2 and HCl or SiH_4 or a hydrocarbon gas to remove subsurface damage produced in the polishing procedure and to obtain a step-and terrace microstructure on the growth surface.

Single polytype replication of a SiC substrate inclined off-axis in the $\langle 11\bar{2}0 \rangle$ direction was achieved via CVD in 6H-SiC thin films in 1987 (Kong *et al.*, 1987; Kuroda *et al.*, 1987) and more fully described in subsequent publications (Kong *et al.*, 1988; Ueda *et al.*, 1990).

Homoepitaxy of 4H-SiC films was subsequently grown using the same approach (Itoh *et al.*, 1994) in the temperature range of 1450–1550°C. The success of the “step-controlled epitaxy” for growth of 4H-SiC and the exceptional electronic and thermal properties of this polytype soon made the former the most common process route and the latter the candidate material of choice for all further growth of films for SiC devices. Optimization of the total growth process including the pretreatment of the polished wafers can result in the deposition of a very flat and smooth homoepitaxial film with few surface features, as shown in Fig. 5.

As indicated in the foregoing paragraphs, high temperatures in the range of 1450–1700°C are required to achieve device quality homoepitaxial thin films of 4H-SiC. As thermal convection is significant at these temperatures, low total growth pressures and/or high carrier gas flow rates are used to achieve a reduction in this phenomenon. The cold wall reactor was the most common system until the mid 1990s. These reactors contained a water-cooled glass or stainless steel double wall annulus that surrounded the SiC-coated graphite sample holder (a.k.a. susceptor) containing the SiC substrate. The sample holder was heated using either an external copper coil powered by a radio frequency generator or an internal graphite resistance heater. However, in this style of reactor, the temperature gradient normal to the wafer surface was very large at these high temperatures, and the resulting large radiation losses caused the heating efficiency to be poor. The large temperature gradient also caused the SiC wafers to warp, the thickness of the deposited films to be non-uniform and the SiC coating on the sample platter to sublime and deposit polycrystalline SiC on the backside of the substrate. The maximum film thickness was $\sim 10 \mu\text{m}$ before the onset of surface roughening that prevented device patterning.

The aforementioned problems with cold-wall reactors were circumvented with the advent and use of the hot-wall (Kordina *et al.*, 1997). This type of reactor normally contains a rectangular SiC- or TaC-coated graphite susceptor that is also the growth chamber. One or more substrates are located on a rotating sample holder at the bottom of the susceptor. The gas stream enters

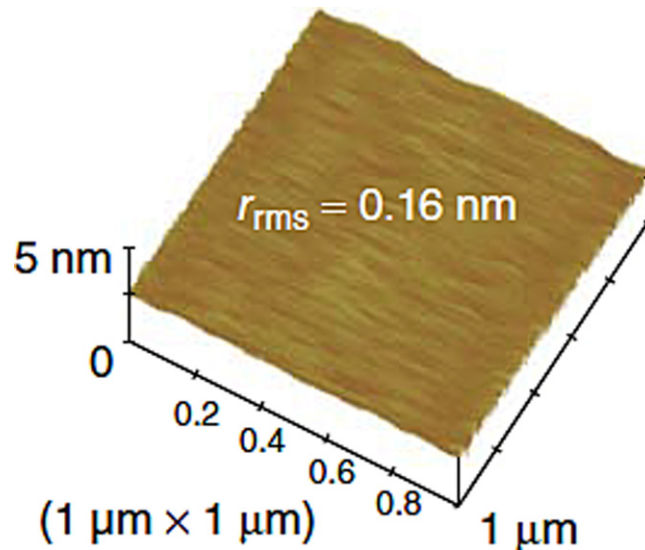


Fig. 5 Atomic force microscopy image of the microstructure of the surface of a homoepitaxial 4H-Si(0001) film (Kimoto and Cooper, 2014).

either parallel or perpendicular to the growth surface of the substrates. Thermal insulation is placed around the susceptor to completely block the emitted radiation to achieve both low energy consumption and uniform thermal fields. This type of reactor has allowed the growth of much thicker ($> 80 \mu\text{m}$) SiC films at faster rates with a much smoother surface microstructure, lower unintentional (background) impurity concentrations than the cold-wall reactors and reduced deposition on the backside the substrates. The warm-wall reactor is a related design wherein the ceiling is heated via radiation from the susceptor. This type of reactor has essentially replaced the cold-wall reactor for the growth of thin films of SiC.

Rapid growth of SiC films in hot-wall reactors must be accompanied by increased supply of reactants into the chamber. However, serious problems often arise with the use of SiH_4 , namely, nucleation of silicon particles and clusters in the gas phase and growth of the 3C polytype on terraces (rather than 4H-SiC from the steps) of the substrate. This has been circumvented by either reducing the total pressure or raising the substrate temperature or by the use of chlorine-containing precursors including either SiCl_4 or one of the chlorosilanes (SiH_xCl_y with x and $y=1, 2$ or 3) without and with a methyl radical attached (eg, SiCH_3Cl_3). These compounds are much more stable to decomposition and less flammable than SiH_4 . Growth of 4H-SiC films at rates of $50\text{--}170 \mu\text{m h}^{-1}$ with good surface microstructure have been reported (Pedersen *et al.*, 2012).

High-Power Devices

Extensive research concerned with the components of device-related processing including doping via ion implantation and subsequent annealing, ion-based dry etching, oxidation to form the native oxide and metallization to form ohmic and rectifying contacts has been conducted over the past 35 years. An excellent review of much of this research has been presented (Kimoto and Cooper, 2014).

At present, most SiC devices have been developed to address three commercial sectors: high-power switching, microwave applications and specialty applications such as sensing and microelectromechanical switching in high temperature, corrosive, toxic and radioactive environments. The aforementioned electronic, thermal and other material characteristics (eg, high breakdown field and the ability to incorporate ionized donor and acceptor atoms) of pure and doped SiC films are most suitable for high-power switching. They are especially applicable above a blocking voltage of 600 V that is the approximate limit for silicon-based power devices, as illustrated in Fig. 6, which provides a comparison of the blocking voltages of Si and SiC unipolar power devices. The useful voltage limit of Si has been reached, as indicated by the blue squares lying essentially on the limit curve for Si. The blocking voltage limit (green) curve for SiC occurs at a much higher range of voltages than Si, and the positions of the green triangles indicate that this latter limit has not been reached for devices in this material. Additional requirements for switching devices are high-power efficiency, high-switching speed and normally off operation. Examples of switching applications that employ voltages exceeding 600 V include (1) DC/DC and DC/AC converters for electric vehicles and hybrid electric vehicles, (2) inverters for photovoltaic energy supplies, industrial motor drives, and uninterrupted power supplies, (3) wind turbine controls and (4) inverters for DC transmission of electric energy. As such, the following paragraphs are devoted exclusively to these devices.

The first SiC-based power device was the Schottky barrier diode which is basically a rectifying metal–semiconductor contact with a low forward voltage drop and a very fast switching action. There is no minority carrier recombination in these devices; thus, the reverse recovery current is zero, as shown by the orange curve in Fig. 7. There is a very small junction-capacitance charge (see dip in orange curve at 10 ns); however, it is negligible in comparison to the equivalent reverse recovery charge in a Si PiN device

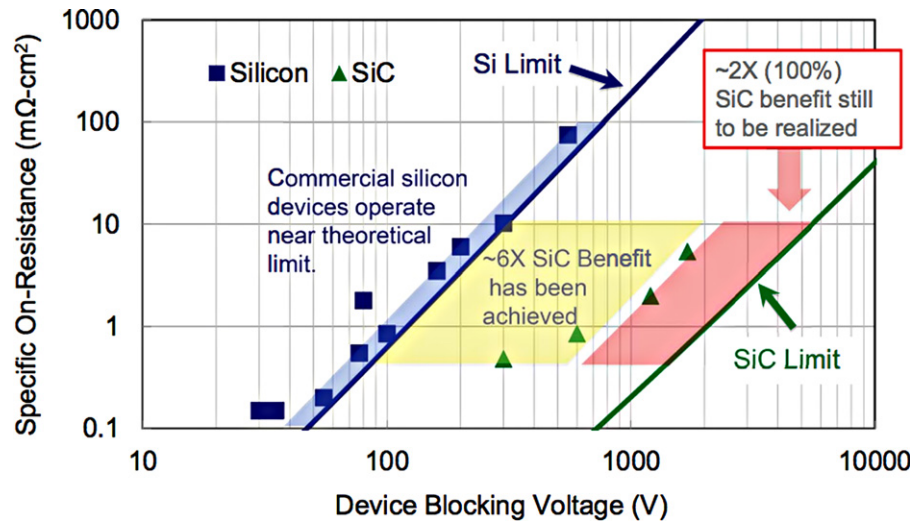


Fig. 6 Comparison of the specific on-resistance as a function of blocking voltage for Si- and SiC-based unipolar power devices (eg, Schottky barrier diodes and MOSFETs). The two curves indicate the limits of blocking voltages for both materials (Courtesy by A. Woodworth).

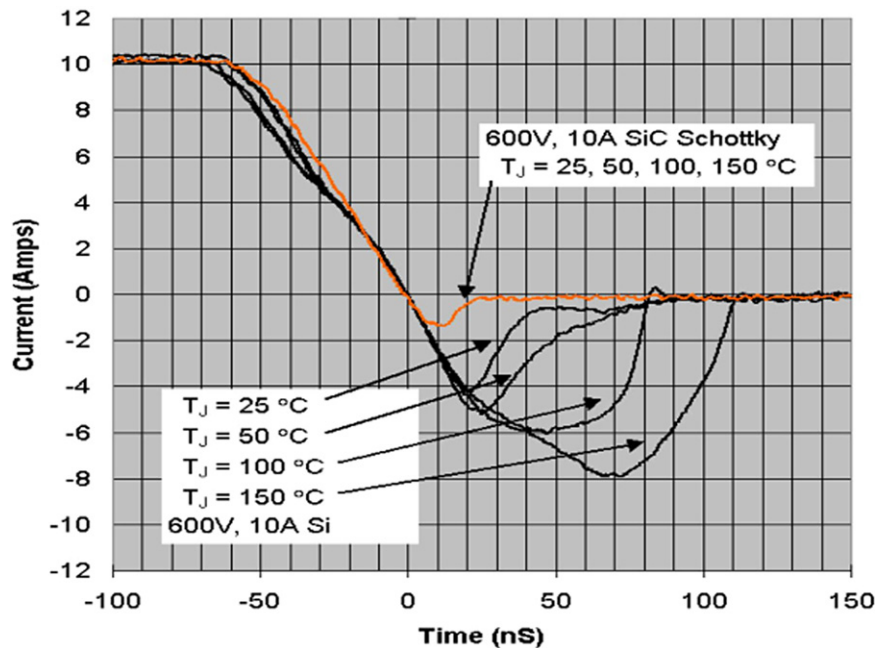


Fig. 7 Reverse recovery currents in a SiC Schottky diode (orange curve) versus silicon PiN junction (black curves) operated at 600 V and 10 A as a function of junction temperature (O'Neill, 2007).

also shown (black curves) in Fig. 7, and it is also independent of temperature, forward current and switching time. SiC SDs also have a zero forward recovery voltage and turn on immediately. These switching characteristics also greatly reduce electromagnetic interference. The fast switching enabled by SiC SDs allows for fabrication of cheaper, smaller and more efficient power systems. For applications that require blocking voltages of up to 1700 V, SiC SDs are the best choice, and their adoption by system designers will continue to grow.

One drawback of the classical SD especially when operated at the higher voltages noted above is that larger electric fields are produced which commonly result in large tunneling currents due to thermionic field emission. A device that combines the advantages of pn junctions and Schottky contacts is the JBS diode, which is also called the pinch rectifier. A schematic of this device is illustrated in Fig. 8. The forward voltage drop is determined by the Schottky junction, whereas, the reverse leakage is limited by the pn junctions. The depletion region of the latter junctions determines the reverse characteristics, if the spacing between these junctions is sufficiently small. The forward characteristics are completely dominated by the Schottky contacts. A closely related device with the same functionalities is the MPS diode.

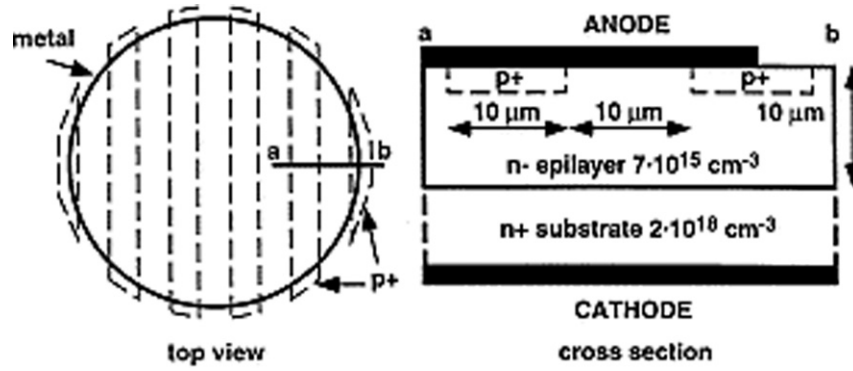


Fig. 8 Schematic top view and cross-section of a 100 μm diameter junction barrier Schottky (JBS) diode (Zetterling *et al.*, 1998).

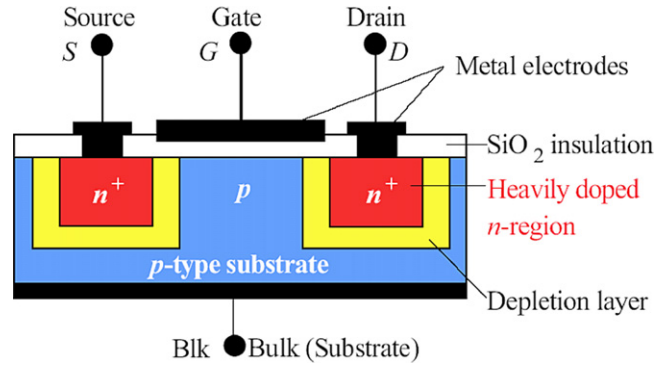


Fig. 9 Schematic of the basic structure of a MOSFET (Kasap, 2005).

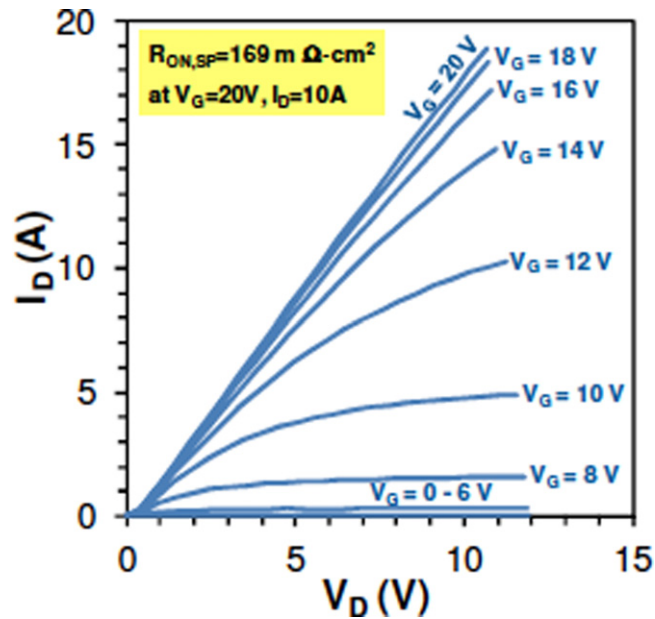


Fig. 10 Forward drain current (I_D)-drain voltage (V_D) characteristics of a 15 kV, 10 A SiC MOSFET tested at room temperature over the gate voltage range of 0–20 V (Cheng *et al.*, 2014).

Though the various types of Schottky barrier diodes were the initial commercial SiC-based power devices, the MOSFET is preferred. (Most of the commercial devices are fabricated using a double diffusion (D) process and are referred to by the abbreviation DMOSFET.) It is used primarily for amplifying or switching electronic signals. The operation of MOSFETs is based on

the effect of an electric field penetrating into a semiconductor. MOSFETs have one input connection referred to as the “gate” which is used to control the flow of electricity between two other connections called the “source” and “drain” shown in Fig. 9. The gate acts as a switch that controls the two outputs. As SiC can be thermally oxidized in the same manner as silicon, it is thus possible to form the entire range of MOS-based devices in this material. MOSFETs provide a smaller on-resistance for a higher blocking voltage than the theoretical limit of Si. In tandem with SiC-based Schottky barrier diodes, they enable switch-mode circuits with improved efficiency for power conversion (AC to DC, DC to AC, DC to DC) and significantly reduced device size. The forward conduction I–V characteristics of a 15 kV and 10 amp-rated SiC MOSFET acquired over the gate voltage range of 0–20 V are shown in Fig. 10. When these devices were switched at ~6 kV at either 20 or 40 kHz, they exhibited a conversion efficiency between 93.2 and 98.5% (Cheng *et al.*, 2014). These results are very promising for high-power and high-frequency applications that can significantly impact the system size, weight and cost of the future advanced medium-voltage systems.

Additional power devices under development in the laboratory are the p-type-insulator-n-type (PiN) diode, the high-voltage IGBT and the gate turn-off (GTO) thyristor. The commercialization of these devices will be governed by the additional reduction in the density of defects and the successful growth of large p-type boules from which p-type substrates can be obtained. For more extensive discussions of the science and technology of SiC-based power semiconductor devices the reader is referred to the following references (Baliga, 2006; Friedrichs *et al.*, 2010; Kimoto and Cooper, 2014; Dimitrijević *et al.*, 2015; Kitabatake and Ohoka, 2015).

Summary

The growth, doping, and device-related science and technology regarding electronic SiC have achieved dramatic progress that has been sufficient to bring large diameter wafers and complex devices made from this material to commercial reality. The emphasis in SiC electronics research and development has been and continues to be driven by the need for devices operative under high-power, high switching frequency and severe environmental conditions. Vapor phase transport is the technique of choice for the production of SiC boules. Wafers trepanned from these boules with finished diameters to 100 and 150 mm are now commercially available. Homoepitaxial growth of very pure and controllably doped thin and thick films of 4H-SiC as the polytype of choice on which to fabricate power devices is now routinely achieved via chemical vapor deposition using various gas-phase chemistries and hot- and warm-wall reactors. Standard SDs, JBS diodes, merged/PiN/Schottky diodes, MOSFETs, IGBTs, GTO thyristors, and PiN diodes in the have been demonstrated 10 to 20 kV classes. The first four types of devices are commercially available at this writing. SiC devices possess the potential to have a significant impact on the system size, weight, high-temperature reliability and cost of modern medium-voltage systems including variable speed drives for electric motors and the integration of renewable energy components including energy storage, micro-grids, and compact pulsed power systems.

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